

MASTAAN UPPAL

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SUMMARY

Engineering Science (Electronics) student at SFU with hands-on experience assembling and validating PCBs, designing digital systems in VHDL on FPGAs, and fabricating electromechanical prototypes. Field Engineer co-op on the 16 km Surrey–Langley SkyTrain expansion. Seeking internships in electronics, embedded systems and semiconductors.

EDUCATION

BASc, Engineering Science — Electronics

Sep 2024 – Present

Simon Fraser University · Surrey / Burnaby, B.C.

- Relevant coursework: ENSC 220 Circuits, ENSC 251 C++, ENSC 252 Digital Logic (VHDL / FPGA), ENSC 280 Statistics, MATH 254 Complex Analysis & Vector Calculus, ENSC 316 Electromagnetics I, ENSC 380 Linear Systems

EXPERIENCE

Field Engineer (Co-op)

Sep 2025 – Apr 2026

Flatiron Dragados · Richmond, B.C. · Surrey–Langley SkyTrain expansion

- Coordinated electrical and roadway IFC design on the 16 km Surrey–Langley SkyTrain expansion, producing detailed Bluebeam markups and supporting RFI submissions that resolved field and design discrepancies.
- Processed survey data with Propeller Aero and CAD tools to analyze site conditions and optimize curb, gutter and roadway layouts, improving constructability for field crews.
- Built Excel-based tracking systems with automated formulas that streamlined quantity tracking, scheduling and field reporting.
- Managed coordination across 5 subcontractors with >\$20M contracts each, supporting progress tracking and alignment with project deliverables.

Engineering Assistant — Marine Design

Jun 2024 – Sep 2024

Autoship Systems Corporation · Vancouver, B.C.

- Used AutoCAD to trace and modify ship design schematics for naval architecture projects.
- Prepared accurate design layers and technical drawings supporting senior engineers.
- Interpreted complex marine blueprints and industry design standards.

PROJECTS

Digital Alarm Clock in VHDL — DE10-Standard (Cyclone V)

ENSC 252 final project · VHDL-2008, Quartus Prime, ModelSim, GHDL, SignalTap

- Designed cascaded modulo counters, alarm registers, an equality comparator and a Moore FSM; simulated in ModelSim/GHDL, synthesized in Quartus and debugged on-chip with SignalTap; successfully demonstrated on hardware.

Custom Motorized Precision Coffee Grinder

Sep – Dec 2024

Electromechanical design · 24 V DC motor, SolidWorks, fabrication

- Designed and fabricated a motorized grinder around a 24 V DC motor and precision burrs; built a custom wood base and mechanical couplings for alignment, prototyped a grind-by-weight system and optimized grind quality through iterative motor-load and durability testing.

PCB Assembly & Functional Validation

Nov – Dec 2024

Electronics · THT + SMT soldering, oscilloscope, multimeter

- Assembled and soldered components using through-hole and surface-mount techniques, interpreted schematics for placement, and verified full board functionality through systematic testing and troubleshooting with oscilloscopes and multimeters.

TECHNICAL SKILLS

Hardware	PCB assembly, THT + SMT soldering, microcontrollers (Arduino), FPGA (DE10-Standard), oscilloscopes, multimeters, DC power supplies, schematic reading, firmware validation
Digital / EDA	VHDL, Quartus Prime, ModelSim, GHDL, SignalTap, EAGLE PCB design, LTspice
Programming	C++, Python, Java, MATLAB, JavaScript, unit testing, debugging, Excel automation
Design / CAD	AutoCAD, SolidWorks, Bluebeam, Propeller Aero, technical drawings